

Solve the following system:

E-Vox

$$\log_2(x) - \log_2(y) = 2$$

$$x + y = \frac{15}{2}$$

$$\log_2(x) - \log_2(y) = 2 \Rightarrow \log_2\left(\frac{x}{y}\right) = 2$$

$$\text{then: } \frac{\ln\left(\frac{x}{y}\right)}{\ln 2} = 2 \Rightarrow \ln\left(\frac{x}{y}\right) = 2 \ln 2$$

$$e^{\ln\left(\frac{x}{y}\right)} = e^{\ln 4}$$

$$\frac{x}{y} = 4 \Rightarrow x = 4y$$

$$\Rightarrow 4y + y = \frac{15}{2}$$

$$5y = \frac{15}{2}$$

$$10y = 15$$

$$y = \frac{15}{10}$$

$$\boxed{y = \frac{3}{2}}$$

$$\text{then: } x = 4\left(\frac{3}{2}\right)$$

$$x = \frac{12}{2}$$

$$\boxed{x = 6}$$

$$S: \left(6, \frac{3}{2}\right)$$

